The Relationship Between Speech and Expressive Language in Children with Cleft Lip/Palate: A Meta-Analysis

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Introduction

Children with non-syndromic cleft lip and/or palate (NSCL/P) have limited consonant inventories and reduced speech accuracy, both of which impact early word use (Chapman, Hardin-Jones, & Halter, 2003; Hardin-Jones & Chapman, 2014; Scherer, Williams, & Proctor-Williams, 2008; Scherer, Boyce, & Martin, 2013; Kaiser, Scherer, Frey, & Roberts, 2016).

Past research has established that speech articulation skills may influence expressive vocabulary outcomes in children with non-syndromic cleft lip and/or palate (Chapman et. al, 2003), but there is limited information available on the strength and direction of this relationship.

The purpose of the current project is to conduct a meta-analysis of research examining the relationship between speech articulation skills and expressive language development in children with non-syndromic cleft lip and/or palate between birth and age 8 years, 11 months.

1. What is the average correlation between speech articulation skills and expressive language at a single assessment point in children with non-syndromic cleft lip and/or palate (NSCL/P)?
2. What is the average correlation between early speech articulation skills and expressive language skills in children with non-syndromic cleft lip and/or palate (NSCL/P)?

Methods

Data extraction:
Studies were coded for study-level characteristics (e.g., publication year and study location) and all eligible speech and language measures were extracted.
- All groups were entered as a separate effect size
- Pre-intervention data was used for intervention studies
- All applicable time points were entered as a separate effect size for longitudinal studies

Effect sizes were coded for the following:
- Construct
- Sub construct
- Measurement source (parent report, standardized test, language sample)
- Sampling context (single word or connected speech), language modality (expressive or receptive)
- Language domain (vocabulary, morphosyntax, omnibus)
- Cleft type

Effect sizes were calculated for the correlation between the speech articulation measure and the expressive language measure, and the receptive language measure. Vocabulary measures included the following:
- Parent-reported expressive vocabulary
- Direct testing vocabulary measure
- Lexical diversity measure from naturalistic language sampling.

The Campbell Collaboration Effect Size Calculator (Wilson, n.d.) was used to compute the correlation effect size and its variance.

Reliability: Review of articles was conducted by an undergraduate student trained in meta-analytic methods, and 100% of the articles were reviewed by a doctoral student or a post-doctoral associate at the title/abstract and full review level to confirm inclusion in the meta-analysis. 100% reliability of effect size calculation was completed by an undergraduate student trained in meta-analytic methods or a post-doctoral associate.

Analysis:
- Standardized correlation effect size metric: Fisher's z
- Robust Variance Estimation
  - Allows for multiple effect sizes to be included from a single study

Methods

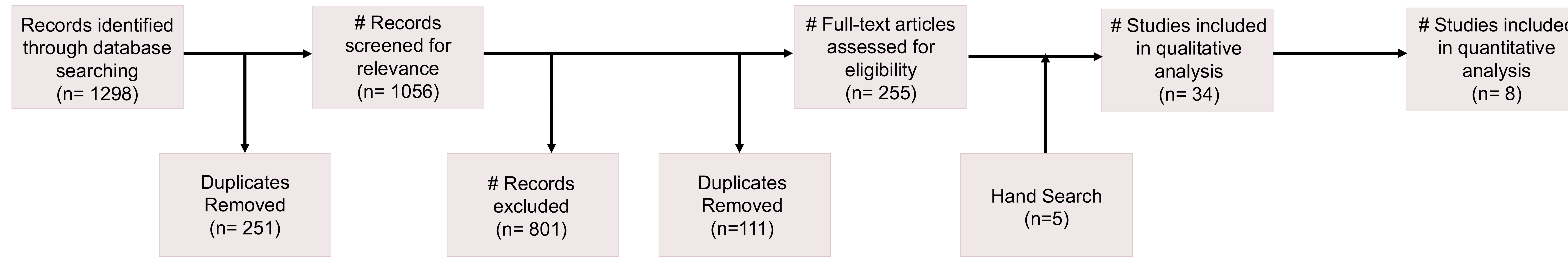

Figure 1. Flow chart of review of search results for inclusion in main project. Only speech and language articles were retained for this presentation.

Search methods: Three databases were searched with 2 parallel searches to find studies for inclusion. The inclusion terms were human participant, cleft, speech, language, reading, and child*. The exclusion terms were: orthopedics, surg*, resonance, adult, adolec*, and otitis media. Title and metadata were extracted for non-duplicate entries and reviewed for age, appropriate topic, and eligible measures. A hand search was completed to identify for inclusion based on a pervious meta-analysis of speech and language abilities in children with NSCL/P (Lancaster, H. S., Lien, K. M., Chow, J., Frey, J., Scherer, N., J., & Kaiser, A., P., in press). For Methods review, articles were screened for study design, non-syndromic, appropriate age, eligible measures, and statistics for calculating effect sizes.

Our results suggest a strong relationship between articulation skills and expressive language development in children with cleft lip/palate, but more data are needed to confirm our preliminary synthesis.

Results

Design	Number of Studies	% Male Mean	Mean N	Mean Age	Mean Age in Months for Primary Palate Repair
Observational	1	61.7	34	18;07	4
Intervention	1	60	30	23	12.47
Longitudinal	6	57.19 [a]	15 [b]	13 [c]	12.69 [d]

Table 1. Descriptive information about the studies included for early speech and language analysis.
Notes. a. Two studies did not report percent male. b. One study's sample changed over time. We used the first time point sample size. c. Calculated using the first time speech was measured. d. Two studies reported more than one left group; therefore we included each group's mean.

Concurrent	n	k	ES_z	SE	t	df	p	95% CI	Tau-sq	I-sq
	5	34	0.93	0.167	5.6	3.74	.006	0.46, 1.41	68.83	0.12

Table 2. Meta-analysis for the concurrent time point effect sizes.

Predictive	n	k	ES_z	SE	t	df	p	95% CI	Tau-sq	I-sq
	4	9	0.195	0.116	1.68	2.98	.193	-0.18, 0.57	0	0

Table 3. Meta-analysis for predictive time point effect sizes.

Conclusions & Clinical Implications

Conclusion:
- Preliminary results show a strong relationship between speech skills and expressive language, but additional analysis may change the strength and direction of the relationship.
- At the same time point, speech and language are highly related to each other; but early speech may not be reliable for predicting later language functioning

Clinical Implications:
- Assess and monitor speech and language skill development simultaneously to better characterize communication skills
- Consider intervention approaches that address early speech skills while also addressing language development

Next Steps:
- Explore avenues to continue acquiring and request speech and language correlation data to add to the sample size and increase statistical power

For More Information, visit: https://tinyurl.com/yb96pbmf

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